Internship in Physical Organic Chemistry

Paper Reading Template

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Fill in one copy of this form for each paper you read. Work through the sections in order. Section A must always be completed. Then fill in **all sections that apply** to your paper — a paper can be both experimental and computational, in which case you complete both Section B and Section D.

Step 1 — Identify the paper type

Read the title and abstract. Then tick **all boxes that apply**.

- ☐ **Review / overview** — summarises a field or a set of methods; does not report new experimental results
- ☐ **Experimental** — reports new measurements, spectra, or observations
- ☐ **Computational / theoretical** — reports calculations, energy profiles, or mechanistic reasoning
- ☐ **Methods** — describes an instrument, technique, or protocol
- ☐ **Mixed** — combines two or more of the above (note which): ____________________________

Section A — Every paper (always complete this)

Bibliographic information

Authors: ____________________

Year: ____________________

Journal: ____________________

Title: ____________________

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What is the main question or topic of this paper?

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List the three most important things this paper tells you.

1. ____________________

   ____

2. ____________________

   ____

3. ____________________

   ____

How will you use this paper in your Background chapter?

- ☐ To introduce the chemical system or compound class
- ☐ To describe the experimental technique
- ☐ To support a specific observation or assignment
- ☐ To introduce a mechanistic argument
- ☐ To provide vocabulary or notation
- ☐ Not sure yet
- ☐ Other: ____________________

Section B — Experimental or Mixed papers

Complete this section if you ticked **Experimental** or **Mixed** in Step 1 and the paper reports measurements or spectra.

What system is studied? (compound, reaction, conditions)

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List three key observations reported in this paper.

Write only what is directly measured or seen — no interpretation yet.

1. ____
   ____
2. ____
   ____
3. ____
   ____

What interpretation do the authors give for these observations?

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Can you clearly distinguish observation from interpretation in this paper?

- ☐ Yes — the paper separates them clearly
- ☐ Partially — some mixing of the two
- ☐ No — the paper presents interpretation as if it were observation
- ☐ Unsure

Notes on figures (which figures are most useful, what do they show?)

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Section C — Review papers only

Complete this section if you ticked **Review / overview** in Step 1.

What vocabulary or concepts does this paper introduce?

List terms you did not know before reading, or terms used differently than you expected.

1. ______________________________
2. ______________________________
3. ______________________________
4. ______________________________
5. ______________________________

Which search terms could you extract from this paper?

These will be used to find additional papers in Google Scholar.

____

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Which references cited in this paper might be worth following up?

Author(s) and year	Why interesting?

Section D — Computational / theoretical papers only

Complete this section if you ticked **Computational / theoretical** in Step 1.

What computational method is used?

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What quantities are calculated?

For each quantity, write what it represents physically.

Quantity	Physical meaning

How do the computational results connect to experimental observations?

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